



**Catastrophic Wildfire Prevention
Consortium (CWPC) at CrowdDoing**

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1. Organization Overview

The **Catastrophic Wildfire Prevention Consortium (CWPC)** at CrowdDoing is a global volunteer-driven initiative dedicated to reducing the risk of catastrophic wildfires through prevention, mitigation, community resilience, and technological innovation. The consortium brings together professionals, researchers, engineers, designers, analysts, and volunteers from diverse disciplines to develop practical solutions that help communities prepare for and reduce wildfire risks. CWPC operates under the broader CrowdDoing initiative, which focuses on solving complex environmental and social challenges through collaborative innovation.

CWPC's mission is to transform the high costs of wildfire inaction into proactive investments in prevention by combining data-driven decision making, technology, community engagement, and cross-sector collaboration. The organization develops digital platforms, decision-support tools, community assessment frameworks, and educational resources that assist governments, nonprofit organizations, researchers, and local communities in improving wildfire resilience.

During my OPT employment, I served as a **Data Analyst** and contributed to multiple cross-functional software and analytics initiatives. My responsibilities included data analysis, software development, visualization, AI-assisted solutions, GIS-based mapping, dashboard development, data engineering, research, and product support. The work directly aligned with my Master's degree in Data Science by applying data analytics, machine learning concepts, database technologies, visualization, and software engineering principles to real-world environmental challenges.

2. Employment Information

Field	Information
Employee Name	Venkata Sri Deepthi Srikotapeetambaram
Position	Data Analyst (Volunteer)
Organization	Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing
Employment Period	August 21, 2025 – May 4, 2026
Supervisor	Claudia Brenner Co-CEO, Catastrophic Wildfire Prevention Consortium (CWPC), CEO, Atomic Strategic Design
Degree	Master of Science in Data Science
University	The University of Texas at Arlington
Work Authorization	F-1 Optional Practical Training (OPT)

3. Reporting Structure

Throughout my employment, I reported directly to:

Claudia Brenner
Co-CEO, Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing
CEO, Atomic Strategic Design

Claudia Brenner provided strategic direction, product guidance, and project requirements across multiple CWPC initiatives. Regular collaboration involved project planning, feature prioritization, research discussions, technical reviews, and progress demonstrations during team meetings.

4. Degree Relevance

Relation of Employment to Master's Degree in Data Science

My employment directly related to my Master of Science in Data Science from The University of Texas at Arlington. Throughout my employment, I applied graduate-level knowledge in:

- Python programming
- Data analysis and statistical modeling
- Machine learning concepts
- Natural Language Processing (NLP)
- Retrieval-Augmented Generation (RAG)
- Dashboard development
- Data visualization
- ETL pipelines
- Geospatial analytics (Mapbox)
- MongoDB
- Streamlit application development
- Research methodology
- Predictive analytics
- Technical documentation

These skills were applied across multiple projects involving AI-assisted donation systems, geospatial visualization, sentiment analysis, wildfire resilience planning, environmental analytics, and dashboard development.

5. Projects Contributed

During my employment period (August 21, 2025 – May 4, 2026), I contributed to the following major initiatives:

1. Donate Feature

Designed and developed data-driven functionality supporting CWPC's online donation platform. Responsibilities included researching nonprofit donation platforms, evaluating payment processing capabilities, analyzing transaction fee structures, studying multi-currency support, designing backend APIs, and developing AI-assisted chatbot capabilities to improve donor engagement and streamline the donation experience.

2. Innovator Map

Developed an interactive geospatial visualization platform that displays wildfire innovation projects, organizations, researchers, and collaborators across the world. The project involved data integration, geographic visualization using Mapbox, data cleaning, spatial analysis, and interactive dashboard development.

3. Community Wildfire Resilience Scorecard

Contributed to the development and enhancement of the Community Wildfire Resilience Scorecard, an assessment framework designed to help communities evaluate wildfire preparedness, identify resilience gaps, and support mitigation planning. Responsibilities included data analysis, dashboard development, visualization, validation, and support for digital implementation of the scorecard framework.

4. Vegetation Management – Community Wildfire Mitigation

Performed large-scale environmental data analysis using wildfire-related datasets to support vegetation management and community wildfire mitigation efforts. Responsibilities included data preprocessing, exploratory data analysis, predictive analytics, GIS visualization, and development of analytical workflows that assisted in identifying wildfire risk factors and prioritizing mitigation activities.

5. Marketing and News Page

Supported the development of CWPC's digital marketing and news platform by organizing content, integrating structured datasets, improving information architecture, and assisting with analytics that enhanced communication of wildfire prevention initiatives and community engagement efforts.

Each of these projects required the application of technical skills acquired through my Master's degree in Data Science, including Python programming, SQL, data visualization, GIS technologies, API integration, MongoDB, machine learning concepts, ETL pipelines, dashboard development, and cloud-based analytical workflows.

6. Project 1: Donate Feature – Analysis & Enhancements

Project Overview

As part of the Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing, I contributed to the research, analysis, and design of an enhanced donation platform for the Prevent Wildfire initiative. The objective of this project was to improve the donor experience by making the donation process more transparent, globally accessible, data-driven, and AI-assisted.

The project focused on analyzing existing donation workflows, identifying opportunities for improvement, and designing technical solutions that would increase donor confidence, reduce payment friction, and support international fundraising efforts. My work combined data analytics, artificial intelligence, dashboard development, payment system research, and user experience design to provide recommendations for future implementation.

The project consisted of four primary components:

- AI-powered Donation Support Chatbot
- Donation Platform Comparative Analytics & Scoring Framework
- Transaction Fee & Multi-Currency Analysis
- Donation Analytics Dashboard and System Architecture

This work directly applied concepts learned during my Master's degree in Data Science, including data analysis, machine learning, natural language processing, visualization, database design, and software engineering.

Project Objectives

The primary objectives of this project were to:

- Improve the online donation experience for global donors.
- Reduce friction during the donation process.
- Analyze leading nonprofit donation platforms to identify best practices.
- Design an AI-powered chatbot capable of answering donor questions.
- Research transaction fee optimization strategies.
- Recommend multi-currency support for international donations.
- Develop dashboards that support data-driven decision making.
- Create technical architecture for a scalable donation platform.

My Responsibilities

During this project, I was responsible for:

- Conducting research on online donation platforms and payment gateways.
- Performing comparative analytics on leading fundraising platforms.
- Designing evaluation and scoring frameworks.
- Building analytical dashboards and data visualizations.
- Designing an AI-powered chatbot using Retrieval-Augmented Generation (RAG).
- Creating technical architecture diagrams.
- Analyzing payment processing fees.
- Investigating multi-currency support for international donors.
- Documenting feature recommendations and implementation strategies.
- Preparing reports and presentations for project discussions.

Technical Contributions

1. AI-Powered Donation Support Chatbot

One of my primary contributions was designing an AI-powered chatbot to assist donors throughout the donation process.

The chatbot was designed using a Retrieval-Augmented Generation (RAG) architecture that combines Natural Language Processing (NLP), semantic retrieval, and Large Language Models (LLMs) to provide accurate and context-aware responses.

The chatbot workflow included:

- Intent detection using NLP
- Entity extraction
- Query embedding generation
- Retrieval of relevant documents from a vector database
- Context-aware response generation using an LLM
- Delivery of accurate donor guidance

The chatbot knowledge base was designed to retrieve information from:

- Frequently Asked Questions
- Donation policies
- Payment information
- Transaction fee documentation
- Multi-currency guidance
- Historical question-and-answer datasets

The objective was to reduce donor confusion, improve response accuracy, and provide real-time assistance for payment-related questions. The chatbot architecture and knowledge flow are illustrated in the project design.

2. Donation Platform Comparative Analytics

To identify the most suitable technologies for the Prevent Wildfire donation platform, I conducted a comparative analysis of several widely used nonprofit fundraising platforms.

The evaluation framework considered multiple criteria including:

- Transaction fees
- Supported payment methods
- Global accessibility
- User experience
- Security and compliance
- Integration capabilities
- Scalability
- Analytics support

Using Python and data analysis techniques, I developed a scoring framework that objectively ranked competing platforms and highlighted their strengths and limitations.

This analysis enabled data-driven recommendations for selecting payment platforms that maximize donor convenience while minimizing operational costs.

3. Transaction Fee Optimization

Another significant component of my work involved researching how nonprofit organizations manage payment processing fees.

I analyzed various fee-handling approaches used by established fundraising platforms, including:

- Organization absorbs processing fees
- Automatic fee addition
- Donor opt-in fee coverage
- Split fee model
- Tier-based donation options

After comparing these approaches, I recommended an **opt-in donor fee coverage model** because it provides transparency, allows donors to choose whether to cover processing costs, and helps maximize the amount received by the nonprofit. I also proposed user interface enhancements such as optional fee coverage and donation round-up features to improve donor flexibility while maintaining a simple donation experience.

4. Multi-Currency Support

To improve accessibility for international donors, I researched and designed a scalable multi-currency donation framework.

The proposed system included support for major currencies including:

- USD
- EUR
- GBP
- CAD
- AUD

Key recommendations included:

- Automatic currency detection
- Real-time exchange rate integration
- Currency conversion transparency
- Storage of both local and base (USD) values
- Consistent reporting for financial analytics
- Expandable architecture for future currencies

I also evaluated exchange-rate providers, discussed caching strategies, and documented recommendations for localization, multilingual support, and scalable database design using MongoDB. The implementation plan described currency configuration, payment gateway integration, reporting, and future enhancements.

5. Dashboard Design and Analytics

To support organizational decision making, I designed analytical dashboards that monitor donation activity and platform performance.

The dashboards included visualizations for:

- Total donations
- Active donors
- Conversion rate
- Donation trends
- Geographic donation distribution
- Campaign performance
- Donation categories
- Currency distribution
- Platform comparison
- Donor engagement metrics
- AI chatbot usage

These dashboards provide administrators with actionable insights into fundraising performance and donor behavior while supporting strategic planning.

Technologies Used

Programming Languages

- Python

Data Analytics

- Pandas
- NumPy
- Jupyter Notebook

Machine Learning & AI

- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Natural Language Processing (NLP)

Data Science

- Comparative Analytics
- Statistical Analysis
- Data Visualization

Visualization

- Matplotlib
- Plotly

Architecture & Design

- AI System Design
- Dashboard Design
- Workflow Modeling
- Technical Documentation

Skills Demonstrated

Through this project, I applied several concepts learned during my Master's in Data Science, including:

- Data analytics and exploratory data analysis
- Statistical comparison and scoring models
- Machine learning concepts
- Natural language processing
- Retrieval-Augmented Generation (RAG)
- Dashboard development
- Data visualization
- Technical architecture design
- Database and information modeling
- Payment analytics
- Research methodology
- AI solution design
- Technical documentation

Project Impact

The Donate Feature project provided a comprehensive roadmap for improving the Prevent Wildfire donation platform by integrating artificial intelligence, data analytics, and modern payment technologies. My contributions supported the organization's goal of creating a more transparent, accessible, and data-driven donation experience for a global audience. The project delivered research-backed recommendations for AI-assisted donor support, payment optimization, multi-currency accessibility, comparative platform evaluation, and analytical dashboards, establishing a scalable foundation for future platform enhancements.

7. Project 2: Innovator Map – Interactive Geospatial Visualization Platform

Project Overview

As part of my work at the Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing, I contributed to the development of the **Innovator Map**, an interactive geospatial visualization platform designed to showcase wildfire prevention initiatives, partner organizations, innovators, research projects, and community programs across different geographic locations.

The primary objective of this project was to create a centralized, map-based platform that enables volunteers, donors, researchers, and community members to discover wildfire prevention projects, understand their impact, and explore opportunities for collaboration and financial support.

The Innovator Map combined geospatial analytics, interactive visualization, and location intelligence to transform organizational data into an intuitive decision-support tool.

Project Objectives

The Innovator Map was designed to:

- Visualize wildfire prevention projects geographically.
- Display organizations, innovators, and community initiatives on an interactive map.
- Increase transparency by allowing donors to understand where their contributions are being used.
- Improve public awareness of ongoing wildfire prevention efforts.
- Connect volunteers, donors, researchers, and organizations through location-based information.
- Support data-driven storytelling through geographic visualization.

My Responsibilities

During this project, I was responsible for:

- Collecting and organizing project location data.
- Cleaning and preprocessing geospatial datasets.
- Developing interactive map visualizations using **Mapbox**.
- Designing map layers and interactive markers.
- Structuring project metadata for visualization.

- Integrating project information with user interaction components.
- Creating data visualizations to communicate project impact.
- Supporting integration between the Innovator Map and the AI-powered donation chatbot.
- Documenting visualization workflows and technical implementation.

Technical Contributions

1. Interactive Geospatial Visualization

I developed an interactive visualization using **Mapbox** that displayed wildfire prevention projects and partner organizations on a dynamic world map.

Each location marker represented a specific project or initiative and provided users with detailed information, including:

- Project title
- Organization
- Geographic location
- Project description
- Community impact
- Areas of focus
- Participation opportunities

The map enabled users to explore projects visually rather than searching through static webpages, making information more accessible and engaging.

2. Data Preparation and Geospatial Processing

Before visualization, project information was collected, cleaned, and standardized to ensure consistency across locations.

This involved:

- Data cleaning
- Removing duplicate records
- Formatting geographic coordinates
- Structuring metadata
- Preparing datasets for Mapbox visualization
- Validating location accuracy

These preprocessing steps ensured reliable visualization and improved the overall quality of the interactive map.

3. Integration with the Donation Support Chatbot

One of my key contributions was integrating the Innovator Map concept into the **AI-powered Donation Support Chatbot** developed for the Donate Feature.

Rather than simply answering donor questions, the chatbot was designed to guide users toward specific wildfire prevention initiatives by leveraging geographic information displayed through the Innovator Map.

When users expressed interest in supporting a particular cause, the chatbot could direct them to relevant project locations where they could:

- Explore ongoing wildfire prevention initiatives.
- Learn about organizations working in different regions.
- Understand the environmental and community impact of each project.
- View project descriptions and objectives.
- Identify opportunities to donate or volunteer.
- Discover projects aligned with their interests.

This integration transformed the chatbot into a navigation and discovery tool, helping donors make informed decisions based on transparent, location-specific project information.

By connecting conversational AI with interactive mapping, the platform provided a more engaging and informative donor experience while increasing trust through visibility into how contributions support real-world wildfire prevention efforts.

4. Improving Donor Transparency

A major goal of the Innovator Map was to improve transparency in nonprofit fundraising.

The platform allows donors to move beyond making a generic donation by exploring exactly where and how their contributions could make an impact.

Using the interactive map, users can:

- See where wildfire prevention projects are located.
- Understand the purpose and goals of each initiative.
- Learn about the organizations leading individual projects.

- Explore project-specific activities and expected outcomes.
- Make more informed decisions about which initiatives they wish to support.

This approach strengthens donor confidence by providing a clear connection between financial contributions and the communities, ecosystems, and wildfire mitigation efforts they help sustain.

Technologies Used

- Python
- Pandas
- NumPy
- Jupyter Notebook
- Mapbox
- Geospatial Data Visualization
- Interactive Mapping
- Data Cleaning
- Geographic Information Visualization
- Plotly
- JSON Data Processing

Skills Demonstrated

This project allowed me to apply several concepts from my Master's in Data Science, including:

- Geospatial data analysis
- Interactive visualization
- Data preprocessing
- Geographic information systems (GIS)
- Data storytelling
- Human-centered data visualization
- Location intelligence
- Integration of AI with geospatial applications
- Dashboard and visualization design

Project Impact

The Innovator Map transformed wildfire prevention data into an interactive, location-aware platform that enhanced transparency, collaboration, and community engagement. By combining Mapbox-based geospatial visualization with the AI-powered Donation Support Chatbot, the project enabled donors, volunteers, and researchers to

discover initiatives geographically, understand the impact of individual projects, and make informed decisions about where to contribute their time or financial support. This integration strengthened donor trust by clearly illustrating how contributions support real-world wildfire prevention efforts while making CWPC's initiatives more accessible to a global audience.

8. Project 3: Community Wildfire Resilience Scorecard – Data Science Initiative Planning

Project Overview

As part of my work at the Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing, I contributed to the Community Wildfire Resilience Scorecard initiative by researching, designing, and documenting multiple data science solutions that could strengthen wildfire prevention, community resilience, and post-fire recovery.

The objective of this project was to identify how data science, artificial intelligence, machine learning, predictive analytics, and geospatial analysis could be applied to improve wildfire preparedness and decision-making for communities, emergency responders, utilities, and policymakers.

During this phase of the project, I collaborated with the CWPC team to discuss technical approaches, define analytical frameworks, identify potential datasets, and document implementation strategies. While production implementation did not begin during my OPT period because the required operational datasets were not yet available, I developed detailed concepts and technical plans that could be implemented once data collection was completed.

This work demonstrated the application of advanced data science methodologies to real-world environmental challenges and served as a roadmap for future wildfire resilience analytics.

My Responsibilities

My responsibilities included:

- Researching data science applications for wildfire prevention.
- Identifying machine learning opportunities across wildfire mitigation.
- Designing predictive analytics workflows.
- Defining data requirements for proposed models.
- Discussing implementation strategies with the CWPC team.

- Preparing technical documentation and project concepts.
- Mapping analytical techniques to real-world wildfire challenges.
- Recommending scalable data engineering and visualization approaches.
- Waiting for production datasets before moving to implementation.

Although these initiatives were not implemented during my employment period, the planning and design work established a technical foundation for future development.

Proposed Data Science Initiatives

1. Early Wildfire Risk Score for Every Neighborhood

Project Idea

One proposed initiative was to develop a predictive wildfire risk score for neighborhoods, similar to weather forecasts or air quality indexes. The goal was to assign each geographic area a daily wildfire risk level (Low, Moderate, High, or Extreme) to help communities prepare for changing environmental conditions.

Proposed Technical Approach

I planned to build a predictive machine learning model that would combine:

- Weather observations
- Vegetation dryness
- Historical wildfire incidents
- Terrain characteristics
- Human activity indicators

Feature engineering techniques would transform these variables into a neighborhood-level wildfire probability score.

Expected Output

- Interactive color-coded wildfire risk map
- Neighborhood-level risk scoring
- Decision-support dashboard for emergency planning

Expected Benefits

- Earlier preparedness for fire departments
- Preventive actions by utility companies
- Easy-to-understand public wildfire alerts

2. Smart Sensor Network for Early Smoke and Heat Detection

Project Idea

This proposed project focused on detecting wildfire ignition during its earliest stages using low-cost environmental sensors instead of relying solely on human reporting.

Proposed Technical Approach

I discussed developing a real-time streaming analytics pipeline capable of processing:

- Air quality measurements
- Temperature
- Humidity
- Infrared heat sensors
- Wind sensors

Time-series analysis and anomaly detection algorithms would learn normal environmental behavior and automatically identify unusual smoke or heat patterns.

Expected Benefits

- Earlier wildfire detection
- Faster emergency response
- Smaller fire sizes
- Improved firefighter safety

3. Home and Infrastructure Vulnerability Scoring

Project Idea

Another concept was to develop a wildfire vulnerability score for residential properties and critical infrastructure, similar to a credit score for wildfire resilience.

Proposed Technical Approach

The planned model would evaluate factors such as:

- Building materials
- Roof type
- Distance to vegetation
- Defensible space

Classification algorithms and risk-scoring techniques would estimate wildfire exposure for individual properties.

Expected Benefits

- Prioritize mitigation investments
- Support insurance evaluations
- Assist homeowners in reducing wildfire risk
- Improve grant allocation strategies

4. Post-Fire Recovery Analytics Dashboard

Project Idea

Rather than focusing only on wildfire damage, this initiative proposed measuring how effectively communities recover after a wildfire.

Proposed Technical Approach

I discussed designing an analytics dashboard that would monitor recovery indicators including:

- Power restoration
- Business reopening
- Healthcare accessibility
- Insurance claim progress
- Infrastructure restoration

Trend analysis and KPI tracking would provide decision-makers with visibility into long-term recovery progress.

Expected Benefits

- Identify recovery bottlenecks
- Improve disaster planning
- Measure community resilience

5. Insurance and Mitigation Return-on-Investment (ROI) Model

Project Idea

This project aimed to quantify the financial benefits of wildfire prevention by estimating avoided losses resulting from mitigation investments.

Proposed Technical Approach

The proposed analytical framework would combine:

- Historical wildfire losses
- Insurance claims
- Mitigation investments
- Scenario analysis
- Cost-benefit modeling

The objective was to estimate how prevention investments reduce future wildfire costs.

Expected Benefits

- Support funding decisions
- Justify mitigation investments
- Demonstrate economic value of prevention

6. Unified Wildfire Intelligence Platform

Project Idea

Wildfire information is often distributed across multiple agencies and systems. This initiative proposed integrating these datasets into a single analytical platform.

Proposed Technical Approach

I discussed creating a centralized data platform that would integrate:

- Weather information
- Satellite imagery
- Sensor networks
- Utility data
- Fire department reports
- Community information

ETL pipelines and interactive dashboards would provide a unified view of wildfire conditions.

Expected Benefits

- Faster decision making
- Improved interagency collaboration
- Single source of truth for wildfire data

7. AI-Based Near-Miss Fire Analysis

Project Idea

Many wildfire incidents never become catastrophic. This initiative proposed studying these "near-miss" events to better understand early warning signs.

Proposed Technical Approach

The planned system would analyze:

- Small wildfire incidents
- Fire department reports
- Weather conditions
- Ignition causes

Pattern mining, anomaly detection, and root-cause analysis would identify recurring conditions associated with potential wildfire escalation.

Expected Benefits

- Improve prevention strategies
- Enhance early warning systems
- Learn from low-cost incidents before major disasters occur

8. Wildfire Smoke Health Impact Prediction

Project Idea

This proposed model focused on predicting the health effects of wildfire smoke before severe air quality conditions developed.

Proposed Technical Approach

The framework would combine:

- Air quality sensor data
- Population vulnerability metrics
- Exposure modeling
- Aggregated healthcare trends

Predictive models would estimate which communities are most vulnerable to wildfire smoke exposure.

Expected Benefits

- Earlier public health warnings
- Better hospital preparedness
- Improved protection for vulnerable populations

9. AI-Powered Grant and Funding Optimization

Project Idea

Many communities struggle to identify funding opportunities for wildfire mitigation projects. This initiative proposed using artificial intelligence to simplify grant discovery.

Proposed Technical Approach

The proposed solution would apply:

- Natural Language Processing (NLP)
- Recommendation systems
- Semantic document analysis

The AI system would compare project descriptions with grant requirements to recommend the most appropriate funding opportunities.

Expected Benefits

- Faster grant discovery
- Higher funding success rates
- Reduced administrative workload

10. Community Readiness Score

Project Idea

Infrastructure alone does not determine wildfire preparedness. This initiative proposed measuring community readiness using both survey responses and operational data.

Proposed Technical Approach

I discussed using:

- Preparedness surveys
- Evacuation participation
- Emergency alert engagement
- Behavioral analytics

These metrics would be combined into a community readiness score reflecting how prepared residents are for wildfire events.

Expected Benefits

- Measure public preparedness
- Identify education needs
- Improve emergency planning

Technologies Planned

Although implementation had not yet begun, the proposed technical stack included:

- Python
- Pandas
- NumPy
- Scikit-learn
- Machine Learning
- Predictive Modeling
- Time-Series Analysis
- Geospatial Analytics
- GIS
- Satellite Data
- ETL Pipelines
- Dashboard Development
- Data Visualization
- NLP

- Recommendation Systems
- MongoDB
- Cloud-based Data Processing

Project Status

During my OPT employment, these initiatives remained in the **research, planning, and solution design phase**. I actively collaborated with the CWPC team to discuss each concept, identify suitable data sources, define analytical methodologies, and document implementation strategies. Development of production models was intentionally deferred because the required operational datasets had not yet been collected and validated by the organization. Once those datasets become available, the documented designs can serve as the foundation for future implementation.

9. Project 4: Marketing & News Analytics Platform

Project Overview

As part of my work at the Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing, I contributed to the development of a data-driven Marketing and News Analytics platform that helps monitor wildfire-related news, understand public sentiment, and communicate wildfire prevention initiatives more effectively.

The objective of this project was to transform large volumes of wildfire news articles into meaningful insights that support organizational awareness, public engagement, and strategic communication. Rather than manually reviewing hundreds of articles, the platform automatically analyzes news content, classifies sentiment, identifies emerging trends, and presents the results through an interactive dashboard.

I contributed by developing a sentiment analysis pipeline in Python and building an interactive Streamlit dashboard to visualize key analytics and insights.

Project Objectives

The primary objectives of this project were to:

- Aggregate wildfire-related news from multiple sources.
- Analyze public sentiment surrounding wildfire events.
- Monitor emerging trends and media coverage.
- Support data-driven communication strategies.
- Build an interactive dashboard for stakeholders.
- Improve awareness of wildfire prevention initiatives.

My Responsibilities

During this project, I was responsible for:

- Performing exploratory analysis on wildfire news articles.
- Developing a sentiment analysis workflow using Python.
- Cleaning and preprocessing textual news data.
- Designing data visualizations for sentiment trends.
- Developing an interactive Streamlit dashboard.
- Creating visual analytics for decision support.
- Presenting insights in an intuitive interface for CWPC stakeholders.

Technical Contributions

1. Wildfire News Sentiment Analysis

One of my primary contributions was developing a sentiment analysis pipeline to evaluate the emotional tone of wildfire-related news articles.

The workflow involved:

- Collecting wildfire-related news content.
- Cleaning and preprocessing textual data.
- Removing stop words and unnecessary formatting.
- Performing sentiment classification.
- Categorizing articles into Positive, Neutral, and Negative sentiment.
- Generating summary statistics and visualizations.

The purpose of this analysis was to understand how wildfire prevention efforts, disasters, environmental policies, and community initiatives were represented in news coverage. These insights can help organizations identify communication trends, measure public perception, and better understand how wildfire-related events are discussed over time. Sentiment analysis is a widely used NLP technique for extracting patterns from large collections of text and supporting communication strategy.

2. Data Processing and Analytics

To prepare the news data for analysis, I developed a preprocessing pipeline that included:

- Data cleaning
- Text normalization
- Tokenization
- Removal of duplicate records
- Missing value handling
- Feature extraction for sentiment analysis

After preprocessing, I performed exploratory data analysis to understand:

- Distribution of news sentiment
- Frequency of wildfire-related topics
- Overall media trends
- Article-level statistics

This transformed raw news articles into structured analytical datasets suitable for visualization.

3. Interactive Marketing & News Dashboard

To make the analytical results accessible to non-technical stakeholders, I developed an interactive dashboard using **Streamlit**.

The dashboard was designed to provide a centralized view of wildfire news analytics and included visualizations such as:

- Overall sentiment distribution
- Positive, Neutral, and Negative article counts
- News trend summaries
- Interactive charts and graphs
- Analytical metrics for communication monitoring

The dashboard enables users to explore wildfire-related media coverage through an intuitive web interface, making it easier to identify communication patterns and understand how wildfire topics are being discussed.

4. Data Visualization

I designed multiple visualizations that transformed analytical results into easy-to-understand insights.

These included:

- Bar charts
- Pie charts
- Trend visualizations
- Summary statistics
- Interactive dashboard components

These visualizations help stakeholders quickly interpret sentiment trends without requiring technical knowledge of the underlying analysis.

Technologies Used

Programming Languages

- Python

Data Science & Analytics

- Pandas
- NumPy
- Exploratory Data Analysis (EDA)

Natural Language Processing

- Sentiment Analysis
- Text Preprocessing
- Tokenization
- Text Cleaning

Visualization

- Matplotlib
- Plotly
- Streamlit

Development Environment

- Jupyter Notebook
- Streamlit

Skills Demonstrated

Through this project, I applied concepts from my Master's in Data Science, including:

- Natural Language Processing (NLP)
- Sentiment Analysis
- Data Cleaning and Preprocessing
- Exploratory Data Analysis
- Data Visualization
- Dashboard Development
- Python Programming
- Analytical Reporting
- Interactive Web Applications

Project Impact

The Marketing & News Analytics project provided CWPC with a data-driven approach to understanding wildfire-related media coverage. By combining sentiment analysis with an interactive Streamlit dashboard, the platform transformed unstructured news articles into actionable insights that could support communication strategies, stakeholder engagement, and public awareness initiatives. Rather than manually reviewing large volumes of articles, users can monitor sentiment trends, explore key metrics, and quickly understand how wildfire prevention topics are being represented across news sources. This project demonstrated the practical application of Natural Language Processing, data visualization, and dashboard development to improve organizational decision-making and communication efforts.

10. Vegetation Management – Community Wildfire Mitigation

Project Overview

As part of my work at the Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing, I contributed to the early stages of the **Vegetation Management – Community Wildfire Mitigation** project. This initiative aims to leverage environmental data, geospatial analytics, satellite imagery, sensor networks, and predictive data science to support proactive wildfire prevention and vegetation management.

Unlike the other projects I worked on, this initiative represents a large-scale, long-term data science effort involving multiple interconnected data sources, including satellite imagery, environmental sensors, weather observations, vegetation conditions, terrain information, and wildfire history. Due to the complexity of integrating these heterogeneous datasets, the project remained in the research and exploratory analysis phase during my OPT employment.

My primary responsibility was to conduct initial analyses, evaluate potential data sources, study wildfire monitoring technologies, and help define the analytical framework that would guide future model development once sufficient operational data became available.

Project Objectives

The primary objectives of this project were to:

- Analyze vegetation conditions associated with wildfire risk.
- Investigate data sources for early wildfire detection.
- Study the integration of satellite imagery and ground-based sensor data.
- Support community wildfire mitigation planning through data analysis.
- Develop a scalable foundation for future predictive wildfire analytics.
- Identify opportunities for proactive vegetation management using environmental data.

My Responsibilities

During this project, I was responsible for:

- Conducting exploratory analysis of vegetation and wildfire-related datasets.
- Researching satellite imagery used for wildfire monitoring.
- Studying environmental sensor technologies for wildfire detection.
- Cleaning and preparing environmental datasets for analysis.
- Performing exploratory data analysis (EDA).
- Evaluating how multiple data sources could be integrated into predictive workflows.
- Participating in technical discussions regarding future implementation.
- Preparing documentation and analytical recommendations for subsequent development phases.

Technical Contributions

1. Initial Vegetation and Wildfire Risk Analysis

I performed an initial analysis of vegetation-related datasets to understand environmental conditions associated with wildfire risk.

This work included:

- Data cleaning and preprocessing
- Exploratory data analysis
- Reviewing vegetation characteristics
- Examining environmental variables that influence wildfire behavior
- Identifying data quality considerations for future predictive models

The analysis established a baseline understanding of how vegetation and environmental conditions could be incorporated into future wildfire mitigation strategies.

2. Satellite-Based Wildfire Monitoring Research

One component of my work involved researching how satellite imagery is used for wildfire monitoring and early detection.

I examined how satellite platforms can provide information such as:

- Active fire hotspots
- Vegetation health
- Burned areas
- Surface temperature
- Smoke plume observations

Satellite data offers broad geographic coverage and is valuable for monitoring wildfire activity over large regions and identifying changes in vegetation over time. Remote sensing is widely used for monitoring wildfire impacts and vegetation recovery across large landscapes.

3. Evaluation of Environmental Sensor Networks

As part of the project's planning phase, I also investigated how environmental sensor networks could complement satellite observations.

The proposed sensor network included measurements such as:

- Air quality
- Temperature
- Humidity
- Smoke concentration
- Heat signatures
- Wind conditions

Based on my initial research, I concluded that ground-based sensors could provide more frequent, localized observations than satellite imagery alone. While satellites are highly effective for monitoring large geographic areas, sensors can capture environmental changes continuously at specific locations and potentially detect abnormal conditions more quickly between satellite observations. The future vision of the project was therefore to combine both technologies rather than rely exclusively on either one.

4. Future Data Integration Framework

A major focus of this project was planning how diverse environmental datasets could eventually be integrated into a unified analytical pipeline.

The planned data sources included:

- Satellite imagery
- Weather observations
- Vegetation indices
- Environmental sensor measurements
- Topographic information
- Historical wildfire records

These datasets were intended to support future predictive models capable of identifying high-risk areas and assisting community wildfire mitigation planning.

Technologies Used

- Python
- Pandas
- NumPy
- Jupyter Notebook
- Exploratory Data Analysis (EDA)
- GIS Concepts
- Environmental Data Analysis
- Satellite Data Analysis
- Data Visualization
- Geospatial Analytics
- Predictive Analytics (Planning)
- Machine Learning Concepts

Project Status

The Vegetation Management project remained **an active, long-term initiative** during my OPT employment because of the significant amount of environmental data required for comprehensive analysis. The project involves integrating satellite imagery, environmental sensor measurements, weather observations, vegetation data, terrain information, and historical wildfire records into a unified analytical framework. Due to the complexity of collecting, validating, and integrating these datasets, development was intentionally carried out in phases.

During my employment, I completed the initial exploratory analysis, evaluated available satellite and environmental data sources, researched wildfire monitoring technologies,

and collaborated with the team to define the analytical direction for future implementation. The project continues to require additional operational data and ongoing analysis before predictive models and decision-support tools can be fully developed.

11. Predictive Analytics Pipeline – Deployment Planning & Technical Review

Project Overview

As part of my work at the Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing, I participated in discussions and technical reviews related to the deployment architecture of the Predictive Analytics Pipeline. The objective of this initiative was to establish a scalable cloud-native architecture capable of ingesting wildfire-related datasets, processing environmental information, generating machine learning predictions, and exposing those predictions through REST APIs for downstream applications and dashboards.

The deployment architecture combined modern cloud technologies, data engineering workflows, machine learning pipelines, and API services to support future wildfire risk prediction and decision-support systems.

Project Objectives

The primary objectives of this initiative were to:

- Design a scalable cloud-based analytics architecture.
- Automate ingestion of wildfire-related datasets.
- Build an end-to-end ETL workflow.
- Support deployment of predictive machine learning models.
- Deliver predictions through REST APIs.
- Monitor system performance and model quality.
- Create a foundation for future wildfire intelligence applications.

My Responsibilities

During this initiative, I was responsible for:

- Reviewing the proposed deployment architecture.
- Participating in discussions regarding data flow and system components.
- Understanding how machine learning models would move from development to deployment.
- Studying cloud deployment workflows and API integration.
- Providing feedback during technical planning discussions.
- Documenting deployment concepts and architecture for future implementation.

While I did not perform the production deployment myself, I gained practical understanding of how data engineering, machine learning, and cloud infrastructure work together to support large-scale predictive analytics applications.

Technical Understanding

1. End-to-End Data Pipeline

I reviewed the proposed end-to-end architecture for processing wildfire-related data.

The planned workflow included:

- Collecting data from external APIs and other data sources.
- Storing raw datasets in cloud storage.
- Cleaning and transforming data through ETL processes.
- Preparing curated datasets for machine learning.
- Persisting processed data in a PostgreSQL database.

This architecture illustrated how raw environmental data could be transformed into structured datasets suitable for predictive analytics.

2. Machine Learning Pipeline

I studied the proposed workflow for deploying predictive machine learning models.

The architecture described:

- Feature engineering
- Model training
- Model evaluation
- Experiment tracking
- Model versioning
- Deployment of the selected model for prediction services

Understanding this lifecycle helped me connect machine learning development with production deployment practices.

3. REST API Architecture

I reviewed how prediction services would be exposed through REST APIs.

The proposed workflow included:

- Receiving prediction requests
- Processing environmental feature data
- Returning wildfire risk scores
- Categorizing predictions into risk levels
- Supporting integration with dashboards and other applications

This demonstrated how machine learning models can be operationalized for real-time decision support.

4. Cloud Deployment Architecture

Through this project, I gained exposure to a modern cloud-native deployment architecture utilizing:

- Azure Data Factory
- Azure Blob Storage
- Databricks (PySpark)
- PostgreSQL
- FastAPI
- Docker
- Azure Container Apps
- GitHub Actions CI/CD
- Grafana
- Prometheus

I also learned how these technologies interact to support scalable data ingestion, model serving, monitoring, and automated deployment workflows.

Technologies Reviewed

- Microsoft Azure

- Azure Data Factory
- Azure Blob Storage
- Databricks
- PySpark
- PostgreSQL
- FastAPI
- Docker
- GitHub Actions
- REST APIs
- MLflow
- Grafana
- Prometheus
- Machine Learning Deployment
- ETL Pipelines
- Cloud Architecture

Skills Developed

Through participation in this initiative, I strengthened my understanding of:

- Cloud-based data engineering
- ETL architecture
- Machine learning deployment workflows
- REST API design
- Model lifecycle management
- CI/CD concepts
- Cloud monitoring
- Data pipeline architecture
- Technical documentation
- Cross-functional collaboration

Project Impact

This initiative provided valuable exposure to production-oriented machine learning and cloud deployment practices within a wildfire analytics context. By reviewing the deployment architecture and participating in technical discussions, I developed a stronger understanding of how data ingestion, ETL pipelines, machine learning models, APIs, and cloud infrastructure integrate to deliver scalable predictive analytics solutions. The experience complemented my graduate studies in Data Science by connecting analytical model development with real-world deployment and operational considerations.

12. Technologies Used

Throughout my employment at the **Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing**, I applied a broad range of data science, artificial intelligence, software engineering, geospatial analytics, and cloud computing technologies across multiple interdisciplinary projects. These technologies supported research, data analysis, predictive modeling, dashboard development, geographic visualization, natural language processing, environmental analytics, and solution design.

Programming Languages

- Python
- SQL

Data Science & Analytics

- Pandas
- NumPy
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Comparative Analytics
- Predictive Analytics
- Feature Engineering
- Risk Scoring Models
- Data Cleaning and Preprocessing

Artificial Intelligence & Machine Learning

- Machine Learning Concepts
- Natural Language Processing (NLP)
- Retrieval-Augmented Generation (RAG)
- Large Language Models (LLMs)
- Sentiment Analysis
- Recommendation Systems
- Time-Series Analysis
- Classification Models
- Anomaly Detection

Geospatial Analytics & Environmental Data

- Geographic Information Systems (GIS)
- Mapbox
- Spatial Data Visualization
- Satellite Data Analysis
- Environmental Sensor Data Analysis
- Location Intelligence

Dashboard Development & Visualization

- Streamlit
- Plotly
- Matplotlib
- Interactive Dashboards
- KPI Visualization
- Business Intelligence Concepts

Databases & Data Engineering

- MongoDB
- PostgreSQL
- ETL Concepts
- Data Integration
- JSON Data Processing
- API Integration Concepts

Cloud & Deployment Technologies

- Microsoft Azure
- Azure Data Factory
- Azure Blob Storage
- Databricks (PySpark)
- FastAPI
- Docker
- GitHub Actions (CI/CD)
- MLflow
- Grafana
- Prometheus

Software Engineering & Research

- Technical Documentation
- Workflow Modeling
- System Architecture Design
- Solution Design
- Product Research
- Comparative Platform Evaluation
- Research Methodology
- Cross-functional Collaboration

13. Technical Skills

Throughout my employment at the Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing, I developed and applied a diverse set of technical skills across data science, artificial intelligence, geospatial analytics, cloud computing, and software engineering projects. These skills were directly aligned with the knowledge and competencies gained through my Master of Science in Data Science program.

Data Science & Analytics

- Data Analysis
- Exploratory Data Analysis (EDA)
- Data Cleaning and Preprocessing
- Statistical Analysis
- Feature Engineering
- Comparative Analytics
- Predictive Analytics
- Data Interpretation

Artificial Intelligence & Machine Learning

- Machine Learning Concepts
- Natural Language Processing (NLP)
- Retrieval-Augmented Generation (RAG)
- Sentiment Analysis
- Large Language Model (LLM) Integration
- Recommendation System Design
- Risk Scoring Methodologies

Data Visualization & Reporting

- Interactive Dashboard Development
- Data Visualization
- KPI Reporting
- Business Intelligence Reporting
- Geospatial Data Visualization
- Technical Reporting

Geospatial & Environmental Analytics

- Geographic Information Systems (GIS)
- Spatial Data Analysis
- Satellite Data Interpretation
- Environmental Data Analysis
- Wildfire Risk Assessment
- Community Resilience Analysis

Data Engineering

- Data Integration
- ETL Concepts
- Database Design Concepts
- API Integration
- Data Pipeline Understanding
- Structured and Unstructured Data Processing

Cloud Computing & Deployment

- Cloud Architecture Fundamentals
- Machine Learning Deployment Concepts
- REST API Concepts
- CI/CD Workflow Understanding
- Model Lifecycle Management
- System Architecture Review

Software Engineering

- Solution Design
- Workflow Design
- Technical Documentation
- System Analysis
- Research and Technical Evaluation
- Software Development Best Practices

Collaboration & Research

- Cross-functional Collaboration
- Technical Research
- Requirements Analysis
- Problem Solving
- Analytical Thinking
- Solution Evaluation
- Project Documentation
- Technical Communication

14. Conclusion

During my employment with the **Catastrophic Wildfire Prevention Consortium (CWPC) at CrowdDoing** from **August 21, 2025, through May 4, 2026**, I contributed to a diverse portfolio of projects focused on wildfire prevention, community resilience, environmental analytics, artificial intelligence, geospatial visualization, data science, and software engineering.

My work included designing AI-assisted donation solutions, developing interactive geospatial visualization platforms, conducting sentiment analysis on wildfire-related news, creating analytical dashboards, researching predictive wildfire resilience models, performing environmental data analysis, and participating in discussions surrounding cloud-based machine learning deployment and system architecture. Across these initiatives, I applied graduate-level knowledge acquired through my **Master of Science in Data Science at The University of Texas at Arlington**, including programming, data analytics, machine learning concepts, natural language processing, GIS technologies, dashboard development, data engineering, cloud computing concepts, and technical research.

Several projects progressed through research, design, prototype development, and analytical evaluation, while others remained in the planning or exploratory phase due to the scale of the initiatives and the need for additional operational datasets before production implementation. Throughout my employment, I collaborated with multidisciplinary teams to analyze requirements, evaluate technical approaches, document proposed solutions, and contribute to the development of data-driven technologies supporting CWPC's mission of catastrophic wildfire prevention.

Overall, this experience provided meaningful practical application of data science principles to real-world environmental challenges while strengthening my expertise in analytics, artificial intelligence, geospatial technologies, visualization, and software

development. The responsibilities I performed were directly related to my graduate education in Data Science and enabled me to develop both technical and collaborative skills through the design, analysis, and support of innovative solutions for wildfire prevention and community resilience.